

Deliverable D - Technical Writeup

Team: Global Intuit Hackers

1. Problem framing & assumptions violated

We act as a small-business lender choosing whom to fund to **maximize realized portfolio value**, while also forecasting cohort default trajectories (B), answering interventional what-ifs (C), and quantifying uncertainty. The dataset deliberately breaks several standard ML assumptions, and each broke a different part of our pipeline:

- **No positivity / deterministic selection (the dominant trap).** Repayment outcomes exist *only* for prior-approved, matured loans, and prior approval is **deterministic** in `prior_underwriter_score`: declined applicants occupy scores $[0, 0.273]$, approved occupy $[0.273, 1.0]$, with **zero overlap**. The IID/representative-sample assumption fails, and - critically - inverse-propensity reject inference is *mathematically void*: $P(\text{approved}|X)=1$ on every labeled row and $1/0$ in the declined region. We confirmed the cutoff empirically (~ 0.273 , 99.6% threshold accuracy, 0 overlap rows).
- **Outcome leakage.** `default_flag`, `days_to_default`, `days_to_full_repayment`, `repayment_status`, `final_recovered_amount`, `observation_status` are post-outcome and blank in test; we quarantine them as features and use them only to define targets.
- **MNAR missingness.** Bank-feed columns are null exactly when no feed is linked, and that is informative (observed default 19.1% no-feed vs 16.7% with-feed). We add explicit missingness indicators rather than imputing.
- **Self-report inflation.** `stated_*` fields are optimistically biased vs the observed bank-feed counterpart; the stated/observed *gap* is predictive (default 8% $\rightarrow$ 28% across its quintiles) but the stated value itself has ~ 0 causal effect on repayment - central to Deliverable C.
- **Right-censoring + drift.** The labeled default rate drifts **15.2% $\rightarrow$ 21.1%** across the 18-month history; late cohorts are under-observed. B requires survival methods, and calibration must track the recent regime.
- **Planted economics trap.** Taking `final_recovered_amount` as the only recovery implies $LGD \sim 0.91$ and a $\sim 9\%$ break-even, which rejects a wide band of profitable loans. But loans **amortize** via daily ACH draws, so a default at the mean day (~ 43 of 60) has already repaid most principal - true realized $LGD \sim 0.25$.

2. Methodology

Shared pipeline. One feature frame for all deliverables: drop leakage/ids/raw timestamp, add MNAR missingness indicators, keep NaN native (HistGradientBoosting handles it), integer-coded categoricals declared as categorical.

Deliverable A - PD + decision. The PD model is a **calibrated blend, $0.4 \cdot (25\text{-model bootstrap HistGBT}) + 0.6 \cdot (L2 \text{ logistic})$** : the signal is largely additive, so the logistic out-ranks the tree, and the blend lifts honest 5-fold OOF AUC $0.774 \rightarrow 0.777$ (positive in all 5 folds). We keep `prior_underwriter_score` (a strong predictor; it is only a collider for *causal* questions, §3). Isotonic calibration on validation gives OOF ECE 0.0018. The decision uses the **brief's exact amortizing NPV**: $\text{repaid} = F + R \cdot r \cdot T / 365$; default at day $t = F + D \cdot (t^* - 1) + \text{rec} - R$. The measured effective LGD is 0.25, so $\text{break-even PD} = 0.0875 / (0.0875 + 0.25) \sim 0.259$ - cross-validated (train OOF 0.265, validation 0.250 bracket it; not tuned to validation profit, which we showed swings 0.19-0.32 and overfits). We fund the observed-support region only (prior-approved, $PD < 0.259$): the positivity violation makes declined-region funding unverifiable, and under the empirical default floor "gated/trust" expansions fall

below* the conservative book while risking \$59-87M of principal. Conservative expected NPV (~\$6.3M) is invariant to the unobservable.

Deliverable B - cohort trajectory (Markov chain). The default definition is an absorbing Markov chain over delinquency states (3 consecutive / 6 cumulative missed daily draws / positive balance at day 90); `days_to_default` is the observed **absorption day**. We model its discrete-time absorption-time CDF $G(a)$ and factor $CDR(\text{cohort } c, \text{ age } a) = (\text{mean PD of A-approved loans in } c) \times G(a)$ - justified because the default-day *shape* is ~proportional across risk strata while the *level* (PD) differs. This reproduces the empirical trajectory at **MAE 0.0056** with the correct, non-trivial shape (near-zero weeks 1-2 from the 3-miss floor; a plateau weeks 9-12 from the days 61-89 dead zone; the day-90 balance-check jump), beating a naive "scale-the-final-rate" baseline (MAE 0.0122).

Deliverable C - counterfactuals. A backdoor-restricted model-perturbation g-formula (full detail in §3).

How they fit together. One calibrated PD model (A) sets each loan's total default mass; the absorbing-chain timing (B) shapes *when* that mass lands; the backdoor-restricted model (C) answers interventions. A's approved set defines B's cohorts, giving a coherent A->B by construction.

3. Causal reasoning & counterfactual methodology

Observational prediction conditions on everything correlated with default; an **interventional** estimate $P(\text{default} \mid \text{do}(F=v))$ must block backdoor paths. Our C model is a **backdoor-restricted, monotone-constrained model-perturbation g-formula**, built on the same modeling spine as A but on a deliberately different feature frame.

Backdoor / collider removal. We drop the three selection nodes - `prior_underwriter_score` (the deterministic selection score), `prior_decision` (the collider), `prior_approved_amount` (a prior-decision output) - because conditioning on any of them opens the path `score -> [decision] <- x` and biases every interventional estimate. They are valuable *predictors* (kept in A) but invalid for `do()`. The causal frame keeps 37 features at val AUC 0.756 vs A's 0.758 - a negligible cost for backdoor validity.

Self-report trap, enforced structurally. `stated_annual_revenue`, `stated_time_in_business`, and `intended_use_of_funds` are application statements, not manipulable business state; their true causal effect is ~0 (the bank-*observed* counterpart drives repayment). We **exclude** them, so `do(stated_*)` returns *exactly* the baseline PD (verified delta 0.000000). This is stricter and more defensible than a shrink factor.

The g-formula for deterministic descendants. For `do(F=v)` we set the raw parent and **recompute** the engineered `leverage = requested_amount / (observed_monthly_revenue * 12)` (we replace the dataset's source-mixed ratio with a clean observed-only one we control), then re-predict. `requested_amount` is a genuine lever (size->leverage->default), not a zero-effect self-report.

Why a naive multivariate counterfactual is wrong, and our fix. `existing_debt_obligations` has a strongly *positive* raw marginal (default 10.5%->25.9% across quintiles) but a *negative* multivariate partial, because it is collinear with `aggregate_credit_utilization` (corr 0.63). A naive "change the input, re-predict" model therefore returns the **wrong interventional sign** for `do(debt(up))`. We inject the unambiguous economic signs **structurally** - HistGBT `monotonic_cst` plus a sign-bounded logistic arm - on the 15 monotone drivers, fixing the sign for -0.002 AUC. All eight adversarial sign checks pass (`do(utilization/debt/delinquency/overdraft (up)) -> PD(up)`; `do(cash_balance(up)) -> PD(down)`; `do(stated_*) -> 0`).

What we give up (honestly). This is model-perturbation, not a fitted structural causal model: it is causal only under our backdoor-sufficiency and descendant-recompute assumptions. We guarantee the *sign* of each effect, not its *magnitude* - a constrained lever can under-respond, and rather than fabricate magnitude with hand-tuned shrink factors, we accept conservative (sometimes ~0) effects on

weak/collinear drivers. 200 of the 300 queried applicants sit in the never-labeled declined region or have no bank feed, so those counterfactuals are extrapolations - we widen their intervals 2.5x. **Regulator defense:** our stated drivers are observed, monotone, and intuitive (utilization, debt burden, invoice delinquency, cash buffer), never self-reported or selection-inherited.

4. Calibration & uncertainty quantification

- **A (PD intervals).** Isotonic-calibrated on held-out validation; the 90% interval is the bootstrap-ensemble 5th/95th percentile of the PD, validated at **10/10 decile-coverage** with mean width ~ 0.12 . We *rejected* a tighter conformal band: it held 90% coverage in-sample only - honest 5-fold OOF coverage collapsed to 38-50% - so we chose honest width over false tightness.
- **B (trajectory intervals).** Predictive bands combine (1) per-loan PD ensemble uncertainty on the cohort mean, (2) bootstrap of the absorption-time curve, and (3) **binomial sampling of each cohort's approved count** - the correct object for a realized *fraction*, avoiding the $1/\sqrt{n}$ CI collapse. Mean width ~ 0.039 .
- **C (counterfactual intervals).** Bootstrap ensemble of the causal model, widened additively in the never-labeled declined region (positivity-limited, flagged as uncalibrated there).
- **Tradeoff.** Throughout we prioritize *honest* coverage over narrow width, validated out-of-fold (A on held-out validation, calibration cross-fit), because an overconfident band fails S_{cal} worse on the drifting test cohort than an honestly wide one.

5. Limitations & what we'd do differently

- **Declined-region positivity wall.** We cannot verify any funding decision in the never-labeled region, so we abstain. If the scorer credits model-extrapolated declined-region NPV, our conservative book leaves expected upside on the table - a deliberate, risk-adjusted choice, not an oversight (the same expansions go *negative* under the empirical default floor).
- **Capped vs uncapped NPV.** The brief's literal default formula credits late defaults ($t > 60$, 22.5% of defaults) above* full repayment; we cap at the repaid margin (conservative). The uncapped reading would raise break-even to ~ 0.28 and fund more - we would confirm the scorer's convention with the organizer.
- **AUC ceiling.** Honest OOF AUC plateaus at ~ 0.775 ; the signal is largely linear and added model complexity, feature engineering, and alternative GBTs gave no generalizing gain. We are confident this is near the data's information limit, not a modeling shortfall.
- **C is model-perturbation, not full identification.** With another day we would write the explicit DAG, add a no-prior-score A variant for defensibility, and attach sensitivity bounds to the counterfactuals rather than point estimates.